

Erdős Problem #196

EP-196

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Must every permutation of $\mathbb{N}$ contain a monotone 4-term arithmetic progression? In other words, given a permutation x of $\mathbb{N}$ must there be indices with either $i < j < k < l$ or $i > j > k > l$ such that x_i, x_j, x_k, x_l are an arithmetic progression?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>